From cobwebs to web: making archives publicly accessible with (open-source) GeoAI



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#experteninbeweging



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Icebreaker

mini-geoguesser

mini-geoguesser

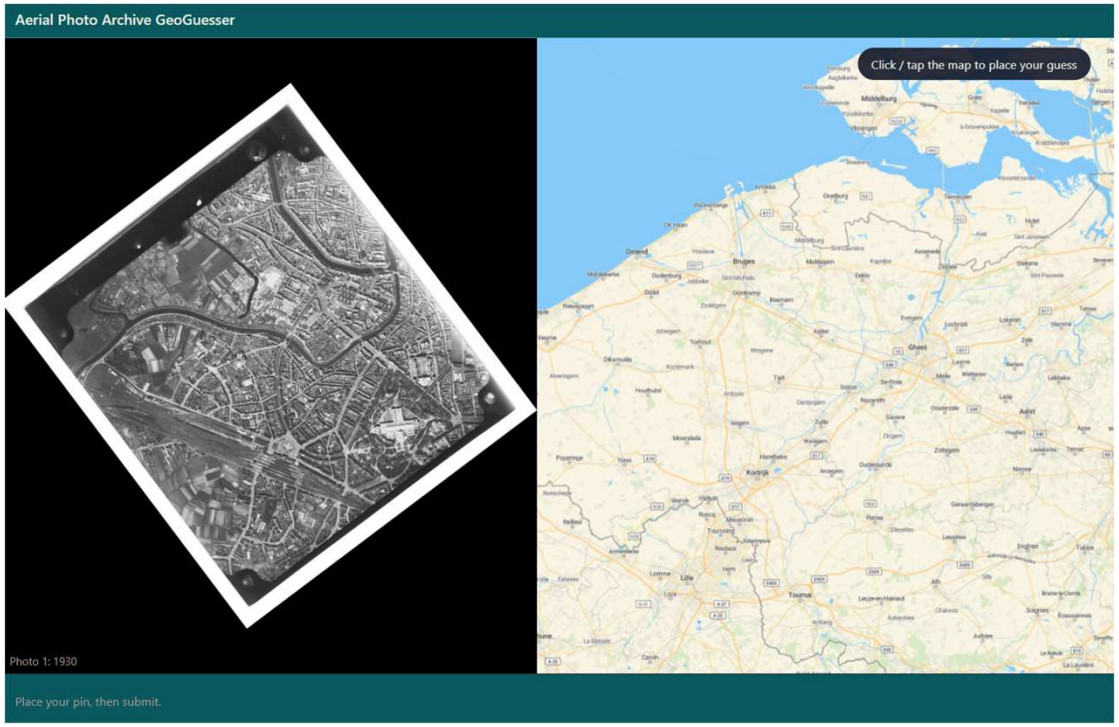


- **scan with phone**

OR

- **https://mow-poc.s3.eu-west-1.amazonaws.com/foss4g/GeoGuess.html**

mini-geoguesser



- Enter your name
- Locate the photo's by placing a pin on the map
- Please stand up when you are finished

Aerial Image Archives

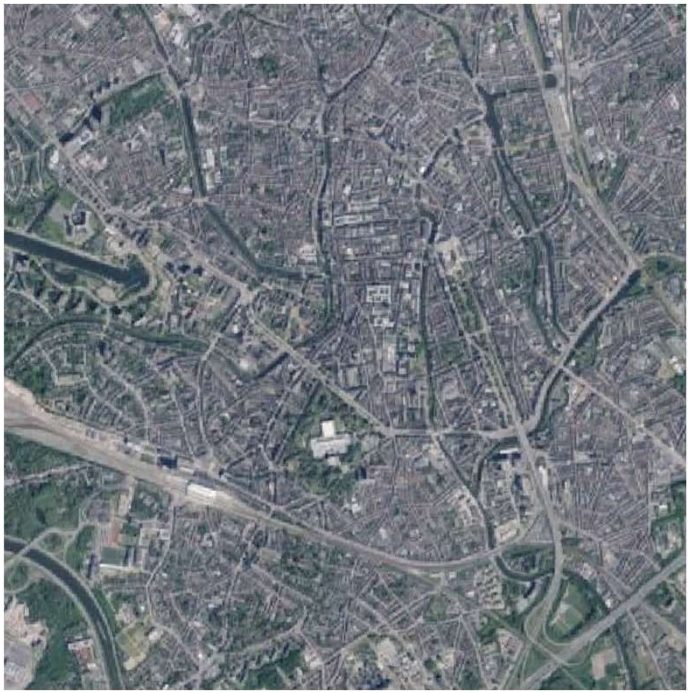


- **63,000+ aerial images**
- **1930-1995**

Aerial Image Archives



Ghent: 1930



Ghent: 2026



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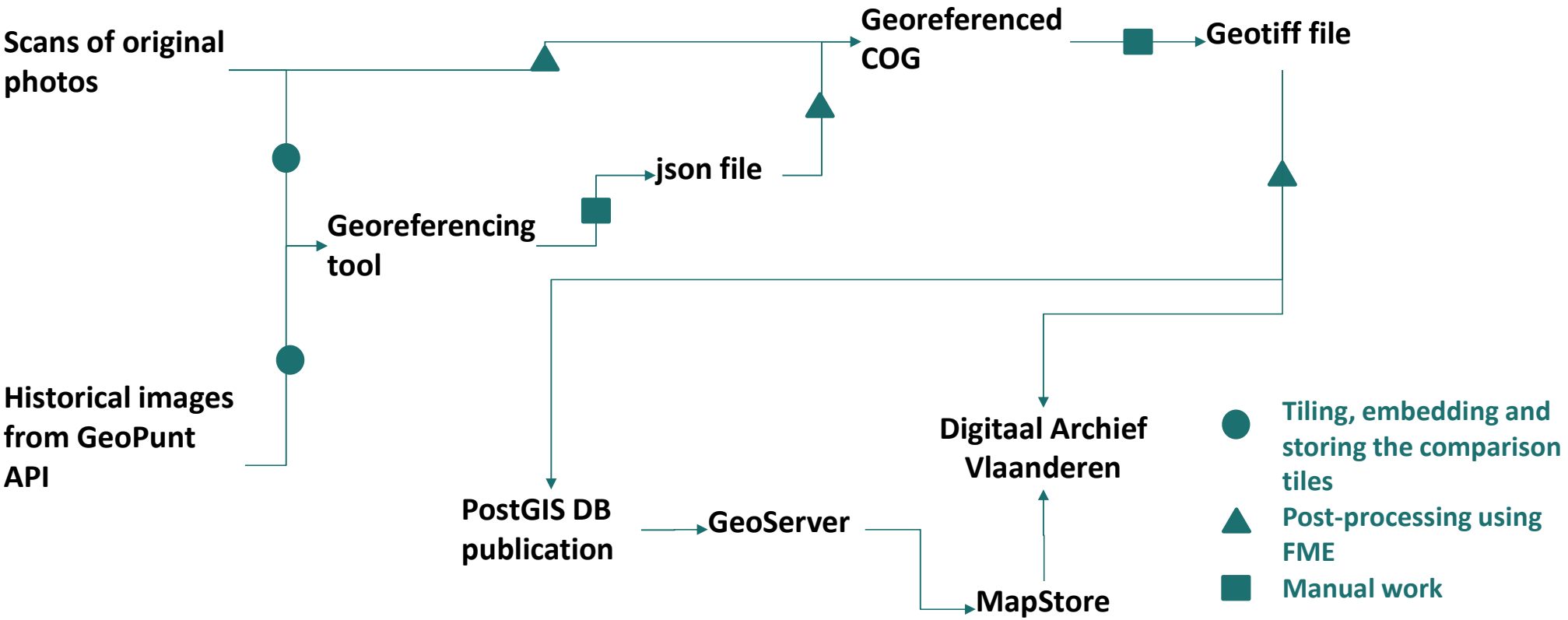
How to georeference historical images?

Georeferencing digitized archival images

- Images are only useful if location information is attached
- Manual process using proprietary software is tedious and slow
- High accuracy preferred



Creating the dataset



Scanning the photos

DigiPost - Digitale post & scanning

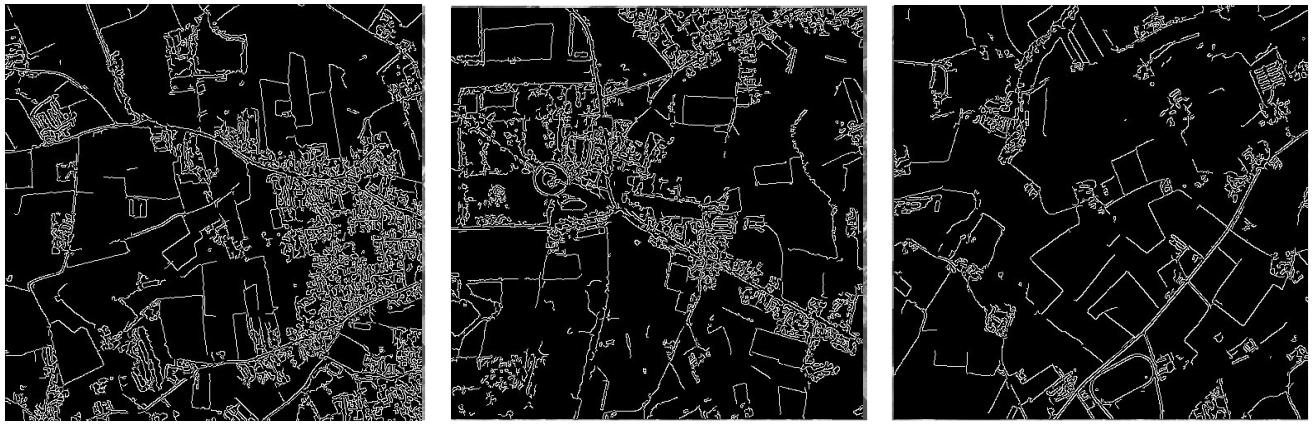


Tiling maps



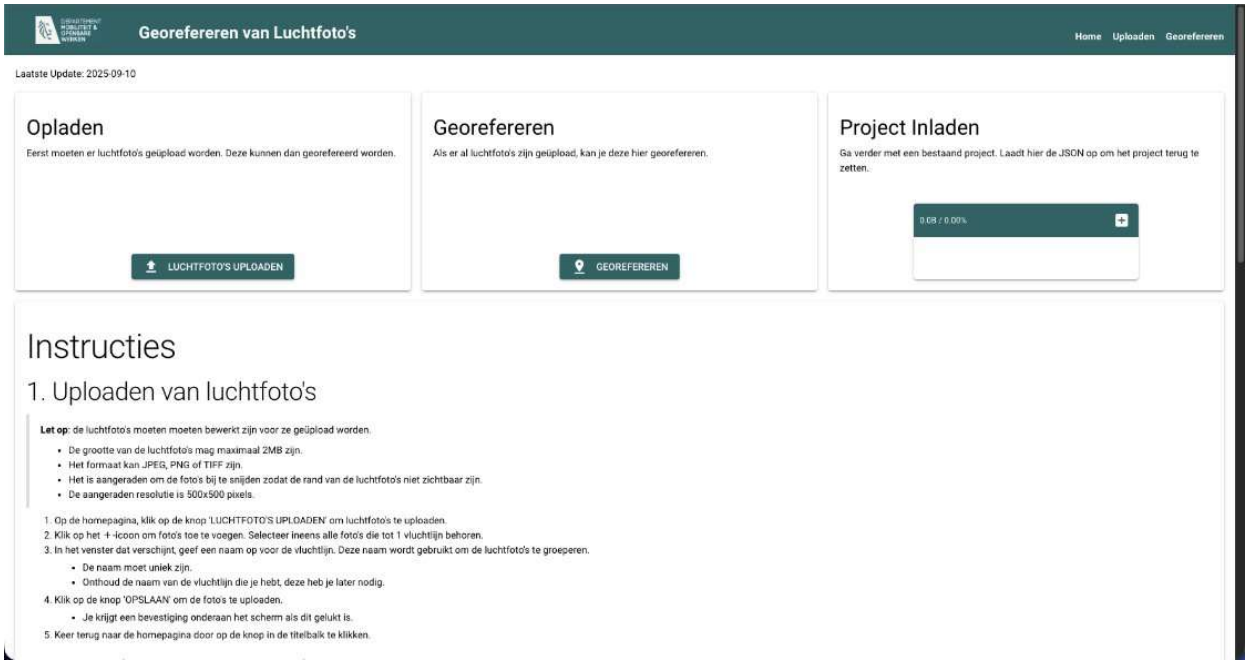
- Geopunt API (geo.api.vlaanderen.be)
- Several different years/layers
- All of Flanders
- 1.5km x 1.5km with overlap

Embedding maps



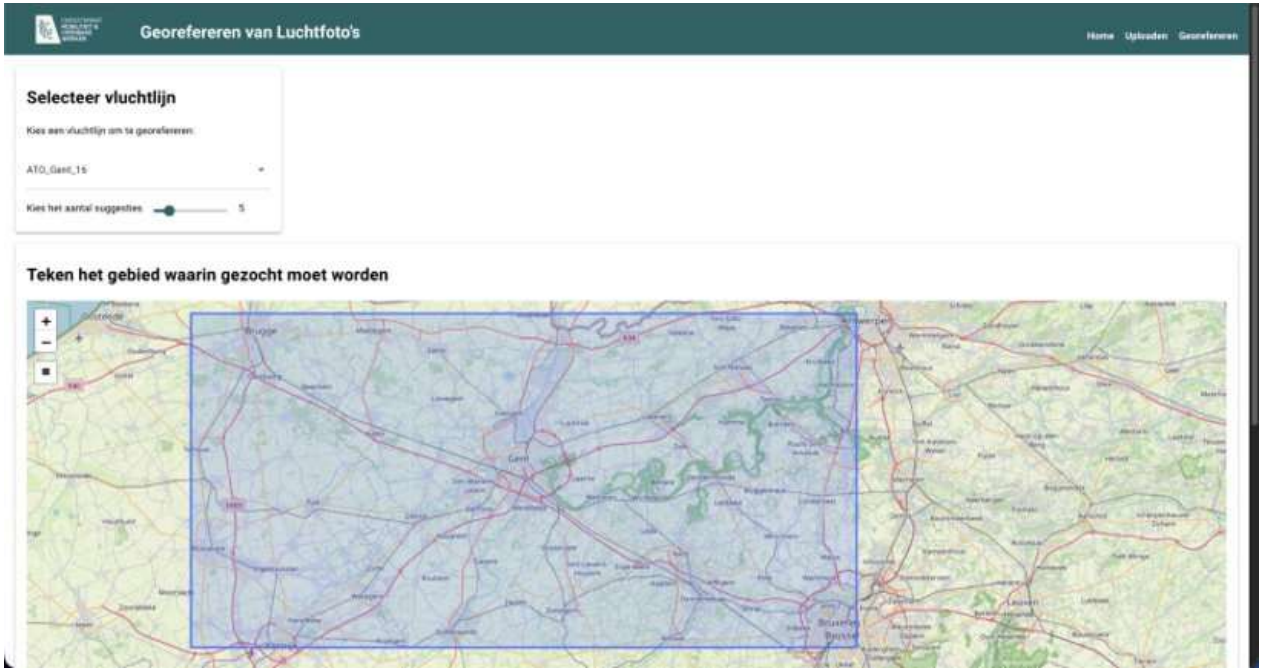
- Edge highlighting preprocessing via Python OpenCV (Open Source Computer Vision Library)
- Embedding using facebook/dinov2-small
- Stored in vector store (eg. CosmosDB)

Manual inference tool



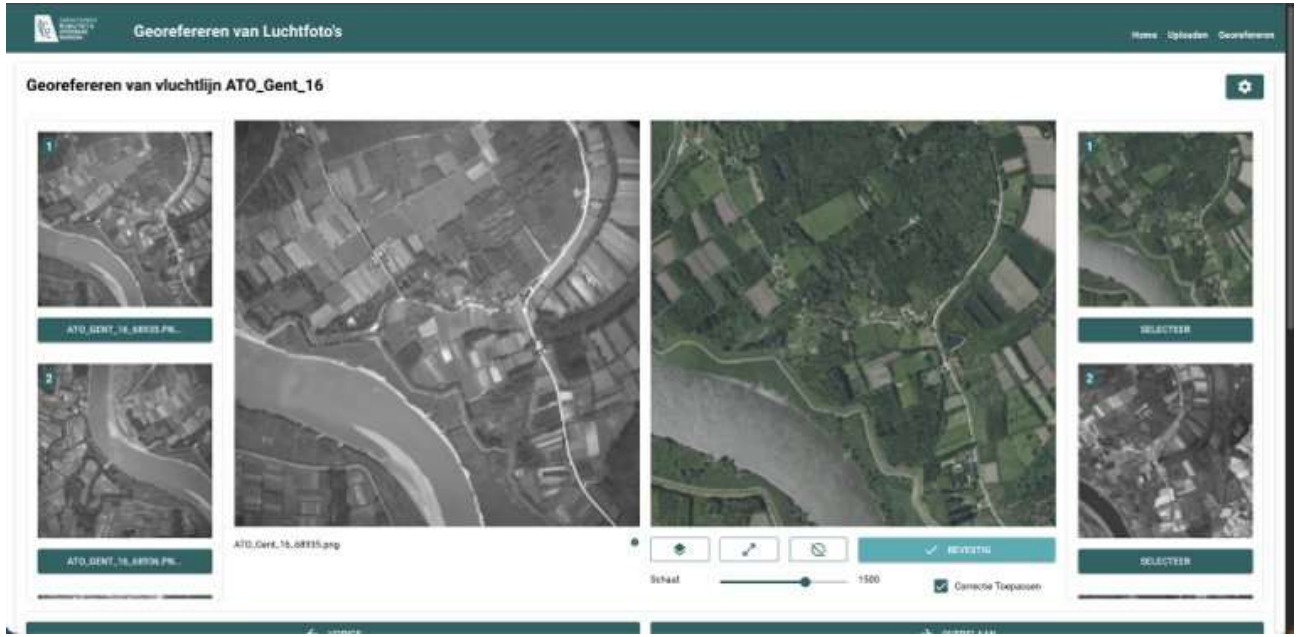
- **Made using NiceGUI**

Manual inference tool



- **Selecting the Search area**

Manual inference tool



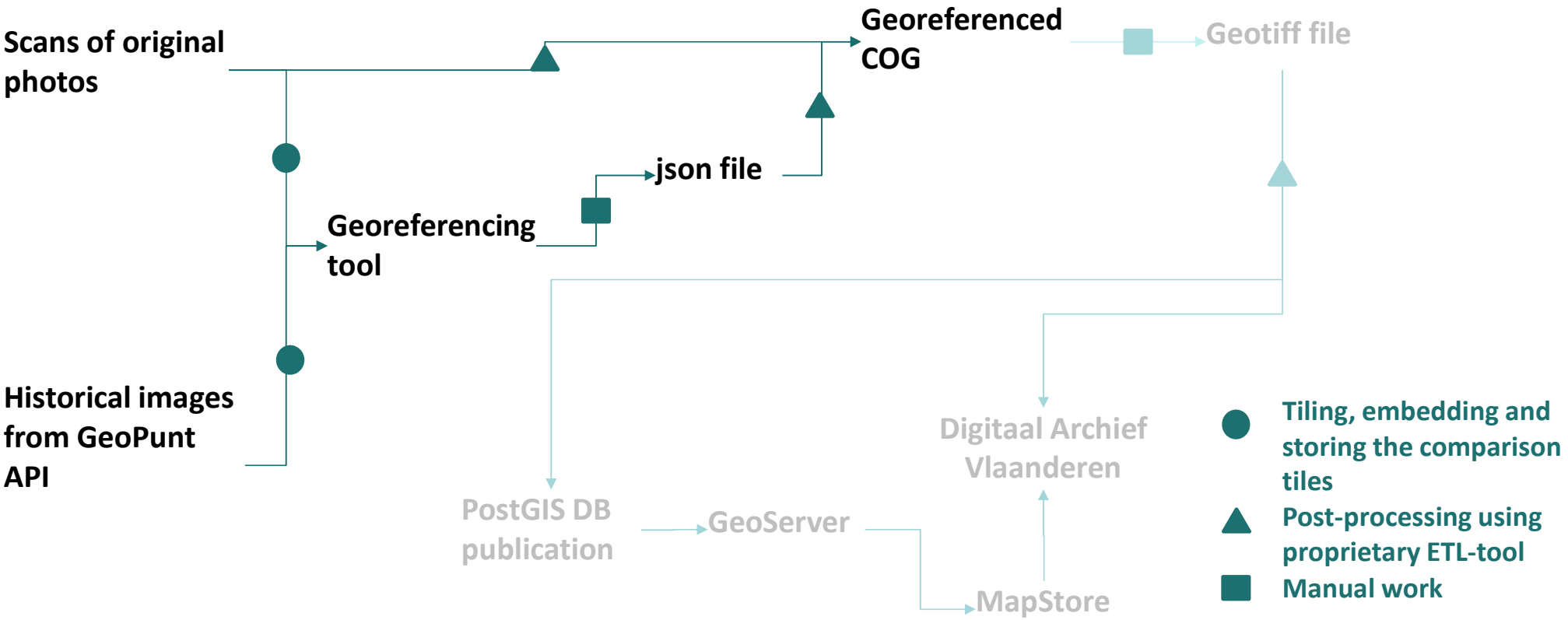
- **User confirms matches**

Manual inference tool



- **Selecting matching landmarks**
- **Optional**

Creating the dataset



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Manual check on map



MapStore Demo

Open MapStore web-GIS viewer:
https://gis-viewer.mow.vlaanderen.be/mapstore/#/context/archief_luchtfotos/22564

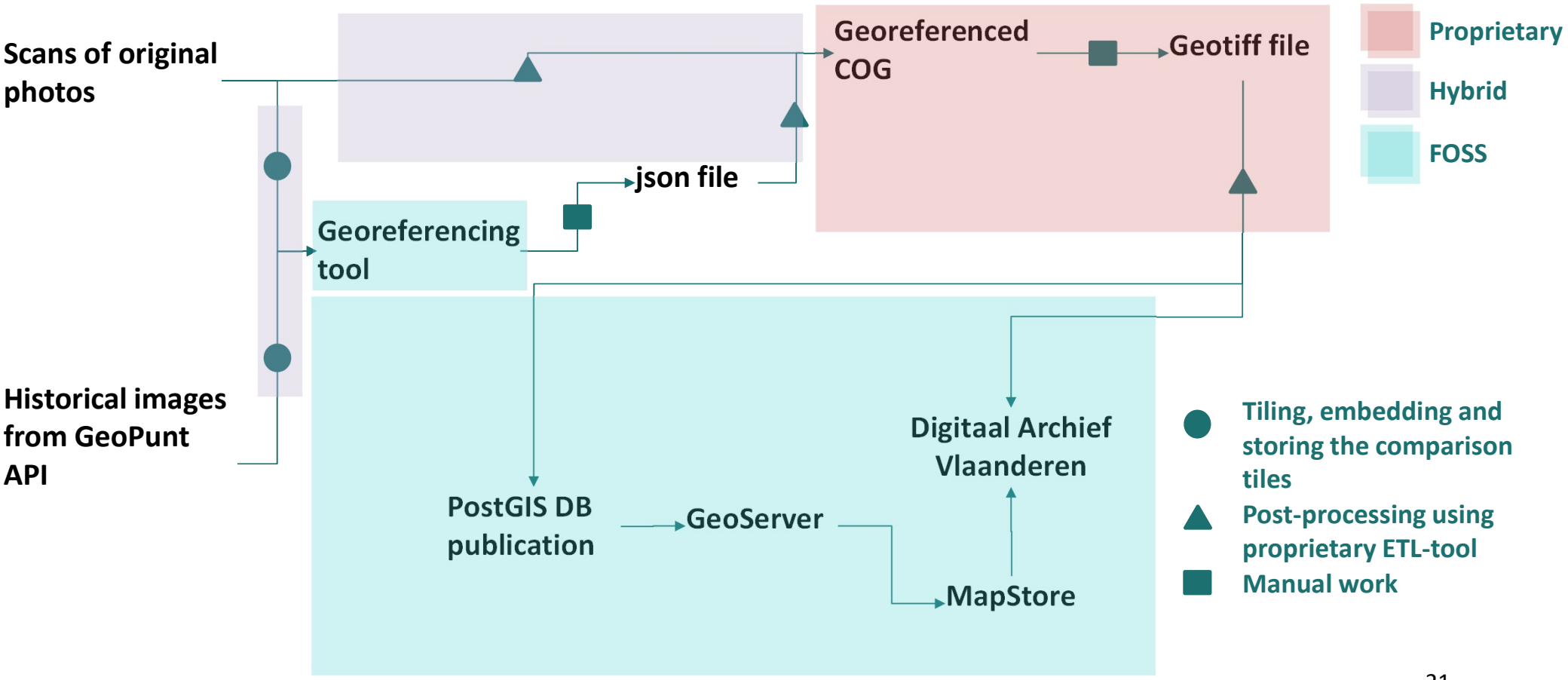


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FOSS or proprietary geospatial software?

a hybrid pipeline

Creating the dataset





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Key Takeaways

1. Vision and strategy FOSS @ DMOW
2. Vision and strategy AI @ DMOW



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Questions?

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